

FINAL PROJECT REVIEW

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

A Salesforce and Agentforce based automation solution for intelligent support-ticket prioritization and assignment.

College	P. B. College of Engineering
Department	B.E. Computer Science and Engineering (Cyber Security)
Academic Year	2026–2027
Project Duration	28 September 2026 – 3 October 2026 (6 calendar days)
Review Type	Final Project Review
Program	Naan Mudhalvan – Salesforce Developer (Skill Wallet)
Mentor	Not assigned in Skill Wallet
Team Lead	Abuthahir A
Team Members	Abinesh M, Chundi Jaghava, Joy Abrahamraj, Guru Saran

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1. Project Overview and User Story

The Customer Support Ticket Priority Prediction and Automated Assignment System uses Salesforce and Agentforce to analyze customer support ticket descriptions, automatically determine the priority as High, Medium or Low, and trigger backend automation for appropriate support handling. The solution uses the **Support Ticket Intelligence** custom object, an **Auto-Launched Flow**, and an Agentforce subagent named **Support Ticket Priority Analysis**.

User Story

A support user provides an Account Name to Agentforce. The system retrieves the latest support ticket of that account, analyzes its description, determines the priority level and returns the priority, the assigned support level and an action message. For a High-priority ticket, the backend Flow creates an urgent handling task and sets the support level to Senior Support Agent.

Core Outcomes

- Automatic classification of support tickets into High, Medium or Low.
- Latest-ticket retrieval based on the customer Account.
- Flow-based task creation for High-priority tickets.
- Consistent support-level assignment for urgent cases.
- Conversational output through Agentforce.

2. Introduction and Purpose

2.1 Project Overview

Customer support teams may receive many tickets and must quickly identify which requests need urgent attention. Manual prioritization can create delays, and important urgency signals in ticket descriptions can be overlooked. This project demonstrates a Salesforce-based approach that combines structured ticket data, Flow automation and Agentforce conversational interaction.

2.2 Purpose

- Automatically classify support tickets as High, Medium or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce manual prioritization and assignment effort.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to analyze ticket details and trigger backend automation.
- Improve the speed and consistency of support-ticket handling.

2.3 Project Objectives

- Create a dedicated Support Ticket Intelligence data model.
- Implement an Auto-Launched Flow for ticket retrieval and priority logic.
- Connect the Flow to an Agentforce subagent.
- Verify the High-priority automation using an actual Salesforce record.
- Document the implementation and evidence for project review.

3. Ideation: Problem Statement and Empathy Map

3.1 Problem Statement

Support teams need a consistent way to analyze the latest ticket for a customer, classify its urgency and start appropriate handling without making the support user perform every backend step manually.

Persona	Need	Problem	Impact
Support Agent	Focus on critical tickets first	Tickets are not always prioritized automatically	Manual prioritization effort
Customer	Get urgent issues resolved quickly	Critical issues may wait behind lower-priority tickets	Concern about response time
Support Manager	Assign tickets to the right support level	Urgency may require manual decisions	Additional coordination effort
Support Agent	Identify urgency from descriptions	Important keywords can be missed	Uncertainty during triage
Support Manager	Reduce SLA-related risk	Older unresolved tickets may remain unnoticed	Potential SLA concern

3.2 Empathy Map Canvas (Persona: Support Agent)

Area	Observation
Think & Feel	Wants critical issues identified quickly so urgent customer problems receive attention.
Hear	Receives customer issue descriptions and support requests.
See	Ticket status, issue type, description, priority level, assignment and SLA risk.
Say & Do	Reviews tickets, handles assigned cases and focuses on urgent support issues.
Pain	Manual prioritization, missed urgency signals and delays in handling critical tickets.
Gain	Automatic priority classification, clearer assignment, urgent-task creation and reduced manual effort.

Design insight: the solution concentrates on three connected needs: reliable ticket retrieval, transparent priority rules and an automated action for High-priority cases.

4. Brainstorming and Idea Selection

Ideas considered: manual ticket registers, spreadsheet-based ticket tracking, a standalone support portal, and Salesforce CRM with Agentforce. **Salesforce CRM with Agentforce** was selected because it provides a structured data model, Flow automation and conversational interaction on one platform.

Capability	How the selected solution addresses it
Automation	Salesforce Flow performs retrieval, decision logic, task creation and output preparation.
AI-assisted interaction	Agentforce provides a conversational interface and invokes the configured backend action.
Priority classification	Description conditions classify tickets as High, Medium or Low.

Capability	How the selected solution addresses it
Urgent handling	High-priority cases create an Urgent Ticket Handling task.
Reduced manual work	The user requests analysis through Agentforce instead of performing each backend step manually.

5. Customer Journey Map

Ticket Creation → Account Identification → Ticket Retrieval → Description Analysis → Priority Classification → Agent Assignment → Urgent Task Creation → Support Handling → Resolution

Stage	What happens
Ticket Creation	Ticket information is stored in the Support Ticket Intelligence object.
Account Identification	The Flow finds the Account using the provided Account Name.
Ticket Retrieval	The latest ticket of that Account is retrieved.
Description Analysis	The description is evaluated against the configured urgency conditions.
Priority Classification	The ticket is classified as High, Medium or Low.
Agent Assignment	High-priority cases get the support level Senior Support Agent.
Urgent Task Creation	A Task is created for High-priority cases.
Support Handling / Resolution	The support team acts on the task and the ticket continues through the normal support lifecycle.

6. Requirement Analysis

6.1 Functional Requirements

FR No.	Functional Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR-6	Support agent / support-level assignment
FR-7	SLA breach risk checking (optional capability)
FR-8	Agentforce conversational analysis

The evidence collected for this review directly demonstrates the core requirements FR-1 to FR-6 and FR-8: the custom object and fields, the Auto-Launched Flow, the Agentforce configuration, the High-priority decision path, task creation and a live conversational test. FR-7 (SLA risk) is an optional capability and is listed under limitations and future scope.

6.2 Non-Functional Requirements

Requirement	Description
Usability	Clear Agentforce interaction and understandable priority / action messages.
Reliability	Consistent classification based on configured conditions.
Performance	Quick retrieval and analysis of the latest support ticket.
Security	Controlled Salesforce access to customer and ticket information.
Maintainability	Flow and Agentforce configuration can be updated as rules evolve.
Scalability	Salesforce automation can support increasing support-ticket volumes.

7. Data Flow and Solution Architecture

The figure shows how a request travels from the support user through Agentforce into the Auto-Launched Flow and back.

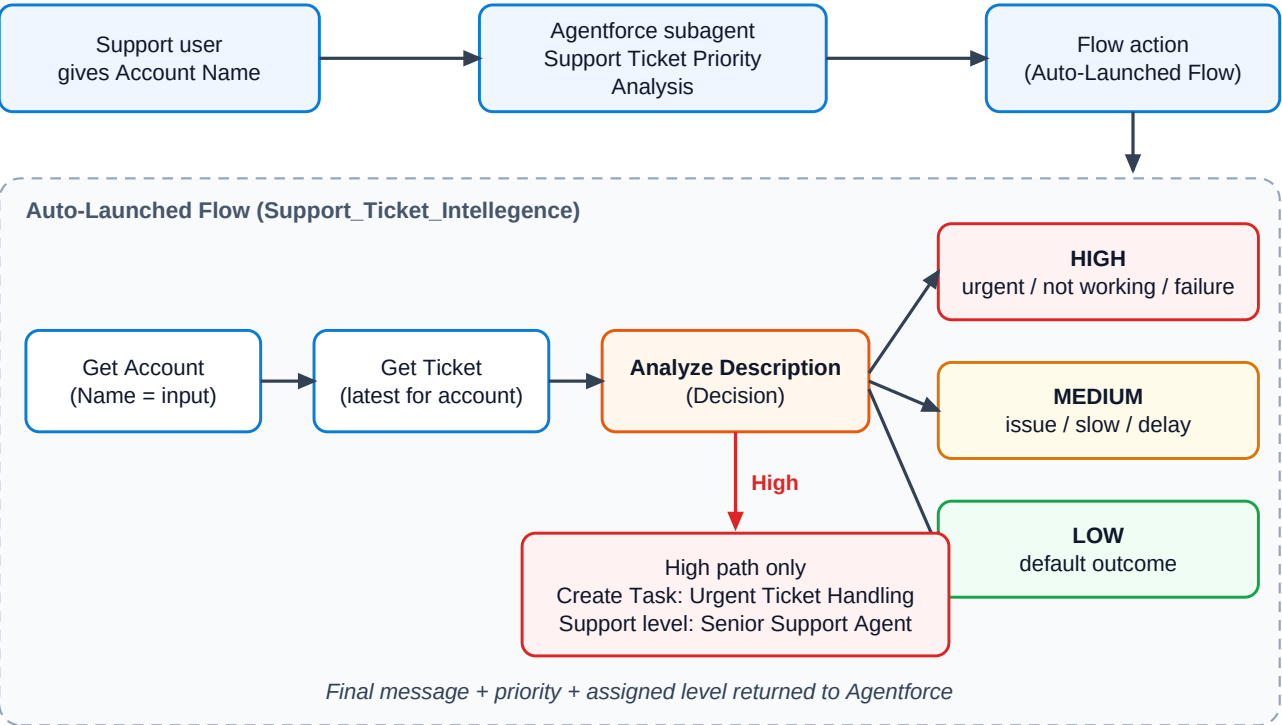


Figure 7.1 – Solution architecture and data flow

Key Data Objects

Object / Record	Role
Account	Identifies the customer account used for ticket retrieval.
Contact	Stores the customer contact associated with the ticket.
Support Ticket Intelligence	Stores ticket-specific information.
User	Supports assignment information.
Task	Records urgent handling work for High-priority tickets.

8. Technology Stack

Layer	Technology / Configuration
Platform	Salesforce Developer Edition
Custom Data Model	Support Ticket Intelligence (Support_Ticket_Intelligence__c)
Related Records	Account, Contact, User, Task
Automation	Auto-Launched Flow (Get Records, Decision, Assignment, Create Records)
AI Agent	Agentforce (Default) agent
Subagent	Support Ticket Priority Analysis
Output	Agentforce conversational response

9. Problem–Solution Fit and Proposed Solution

Problem	Solution	Expected Benefit
Manual ticket prioritization	Flow analyzes the ticket description and sets the priority.	Faster identification of urgent cases.
Urgent issues may be missed	Configured High-priority conditions identify urgent signals.	Critical cases receive attention.
Manual task creation	Flow creates a Task for High priority.	Reduced repetitive work.
Manual support assignment	Flow sets the support level to Senior Support Agent for High priority.	More consistent handling.
Need for conversational access	Agentforce analyzes the ticket through the configured subagent.	Easier user interaction.

Parameter	Project Definition
Problem	Manual ticket prioritization and assignment can delay handling of critical customer support issues.
Solution	Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System.
Innovation	Agentforce conversational analysis combined with an Auto-Launched Flow.
Impact	Faster identification of critical tickets, reduced manual effort and more consistent support handling.
Business Context	Customer support operations involving customers, support agents, support managers and AI-assisted ticket handling.
Scalability	Supports increasing ticket volumes through Salesforce automation, with scope for expanded rules and monitoring.

10. Project Planning, Team Contributions and Timeline

The team allocation below is taken from the Skill Wallet project workspace.

Team Member	Role	Phase 1 Assigned Work
Abinеш M	Member	Understanding Business Requirements
Abuthahir A	Team Lead	Designing Data Model and Security Model
Chundi Jaghava	Member	Defining Project Scope & Objectives
Joy Abrahamraj	Member	Gathering & Analyzing User Needs
Guru Saran	Member	Identifying Key Salesforce Features & Tools Required

Phase 2 – Development contribution of Abuthahir A

Milestone	Assigned Work
Milestone 1	Salesforce Account
Milestone 2	Objects Creation
Milestone 3	Creating a Tab for Support Ticket Intelligence Object

Timeline

Item	Detail
Project Start	28 September 2026
Project Completion	3 October 2026
Duration	6 calendar days
Phases	Phase 1: Requirement Analysis & Planning; Phase 2: Salesforce Development – Backend & Configurations

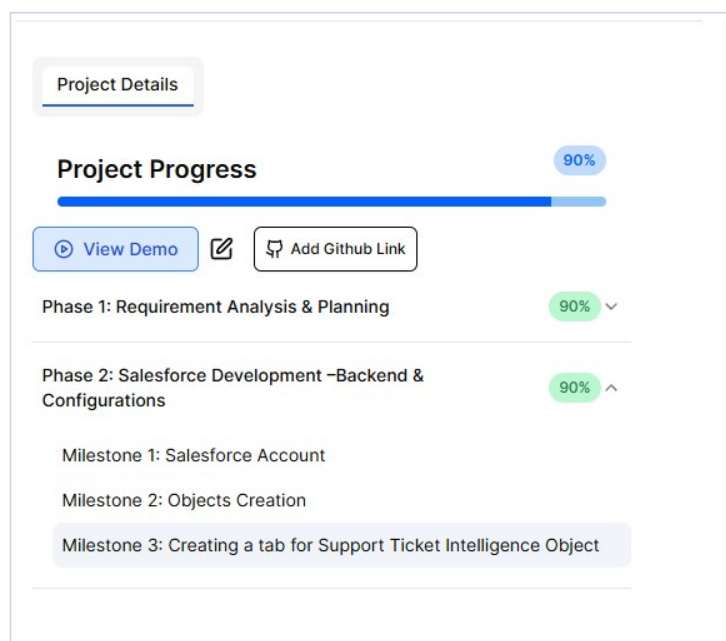


Figure 10.1 – Phase 2 milestones in the Skill Wallet workspace

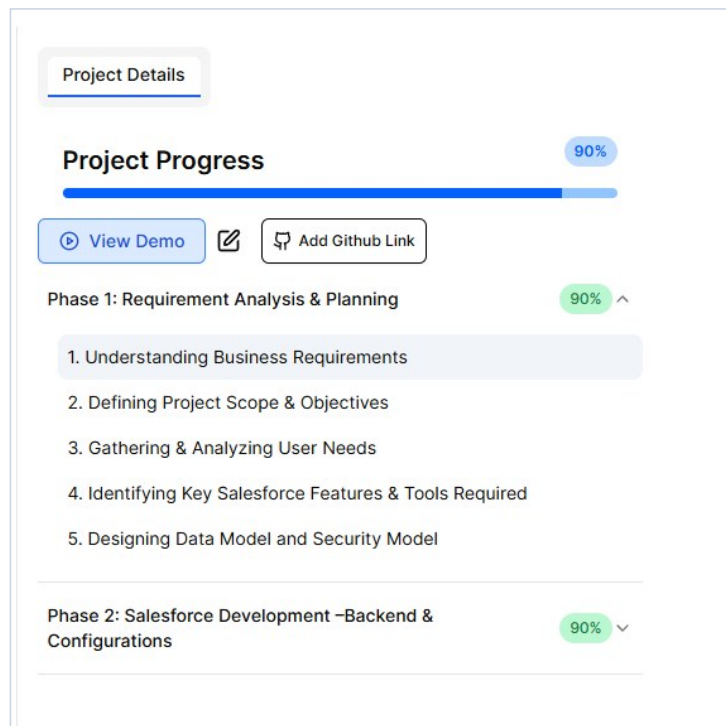


Figure 10.2 – Phase 1 work items and project progress in Skill Wallet

11. Development: Salesforce Account

A Salesforce Developer Edition org was used for the implementation. All configuration (object, Flow and Agentforce) was done in Salesforce Setup and the application environment.

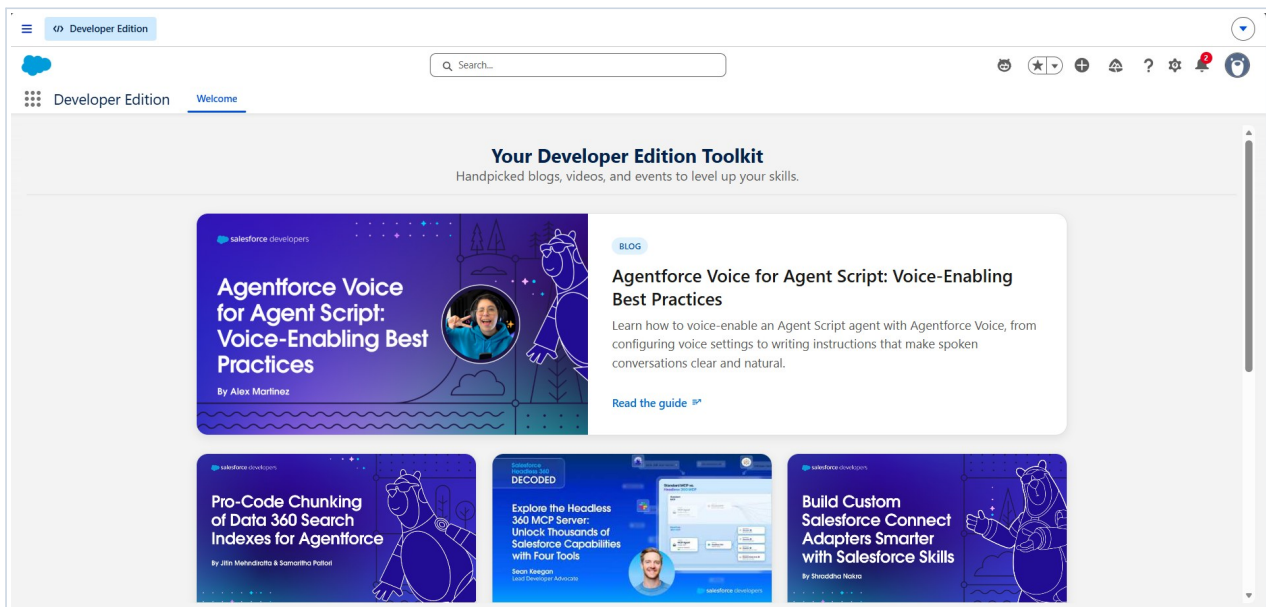


Figure 11.1 – Salesforce Developer Edition environment used for the project

12. Development: Support Ticket Intelligence Object

Custom object: Support Ticket Intelligence | **API name:** Support_Ticket_Intelligence__c

Field	API Name	Type	Purpose
Ticket Number	Ticket_Number__c	Auto Number	Ticket identifier using TKT-{0000}
Customer	Customer__c	Lookup (Account)	Associates the ticket with the customer Account
Contact	Contact__c	Lookup (Contact)	Associates the customer contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
Description	Description__c	Long Text Area	Issue details used for analysis
Priority Level	Priority_Level__c	Picklist	Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Created Date	Created_Date__c	Date	Ticket date
Assigned To	Assigned_To__c	Lookup (User)	Support assignment information
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Optional risk flag
Resolution Time (hrs)	Resolution_Time__c	Number	Time taken for resolution

The Fields & Relationships page also lists Salesforce system fields (Created By, Last Modified By, Owner) and the standard record *Name* field, which is configured as an Auto Number field. Important relationships are Customer → Account, Contact → Contact and Assigned To → User; they allow the Flow to find the latest ticket of a customer and carry assignment information.

Support Ticket Intelligence				
Fields & Relationships				
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Assigned To	Assigned_To__c	Lookup(User)		✓
Contact	Contact__c	Lookup(Contact)		✓
Created By	CreatedById	Lookup(User)		
Created Date	Created_Date__c	Date		
Customer	Customer__c	Lookup(Account)		✓
Description	Description__c	Long Text Area(32768)		
Issue Type	Issue_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User/Group)		✓
Priority Level	Priority_Level__c	Picklist		
Resolution Time (hrs)	Resolution_Time_hrs__c	Number(16, 0)		
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox		
Status	Status__c	Picklist		
Ticket Number	Ticket_Number__c	Auto Number		
Ticket Number	Name	Auto Number		✓

Figure 12.1 – Support Ticket Intelligence: Fields & Relationships

13. Development: Ticket Record

As Team Lead, Abuthahir A completed the Phase 2 milestone of creating a tab for the Support Ticket Intelligence object, which gives a direct entry point to ticket records. The sample record **TKT-0002** belongs to the customer **Abdul** (contact Abdul Rahman). It has issue type Technical, description “urgent” and priority High. The Activity panel shows Urgent Ticket Handling tasks created by the automation.

The screenshot displays the Salesforce Developer Edition interface for the Support Ticket Intelligence object. The record TKT-0002 is shown with the following details:

Field	Value
Ticket Number	TKT-0002
Owner	ABUTHAHIR A
Ticket Number	TKT-0009
Customer	Abdul
Contact	Abdul Rahman
Issue Type	Technical
Description	urgent
Priority Level	High
Status	New
Created Date	10/3/2026
Assigned To	

The Activity panel on the right shows two 'Urgent Ticket Handling' tasks, both with a 'No due date' and a 'Show All Activities' button.

Figure 13.1 – Support Ticket Intelligence record TKT-0002 with Urgent Ticket Handling tasks in Activity

14. Development: Auto-Launched Flow

Flow label: Support_Ticket_Intelligence – V1 (name as saved in the org) | **Type:** Auto-Launched Flow (no trigger, called from Agentforce) | **Status:** Active

14.1 Variables

Label	API Name	Type	Direction
Account Name	varAccountName	Text	Input
Account Id	varAccountId	Text	Output
Ticket Id	varTicketId	Text	Output
Priority Level	varPriorityLevel	Text	Output
Assigned Agent	varAssignedTo	Text	Output
Action Message	varActionMessage	Text	Output

14.2 Flow Elements

Element	Type	Purpose
Get Account	Get Records	Finds the Account whose Name equals varAccountName (first record).
Acc Name	Assignment	Stores the retrieved record values in the output variables.
Get Ticket	Get Records	Finds the latest Support Ticket Intelligence record for the account (Customer__c = account Id, newest first).
Analyze Description	Decision	Evaluates the description and routes to the High, Medium or Low (default) path.
High / Medium / Low Path	Assignment	Sets varPriorityLevel to High, Medium or Low.
Create Task	Create Records	High path only: creates the Urgent Ticket Handling task.
Set Assigned Agent	Assignment	High path only: varAssignedTo = Senior Support Agent.
Set Final / Final Medium / Final Low Message	Assignment	Sets varActionMessage for each priority.

14.3 Priority Rules

Priority	Configured Description Conditions
High	Description contains: urgent, not working, failure
Medium	Description contains: issue, slow, delay
Low	Default outcome when the High and Medium conditions are not met

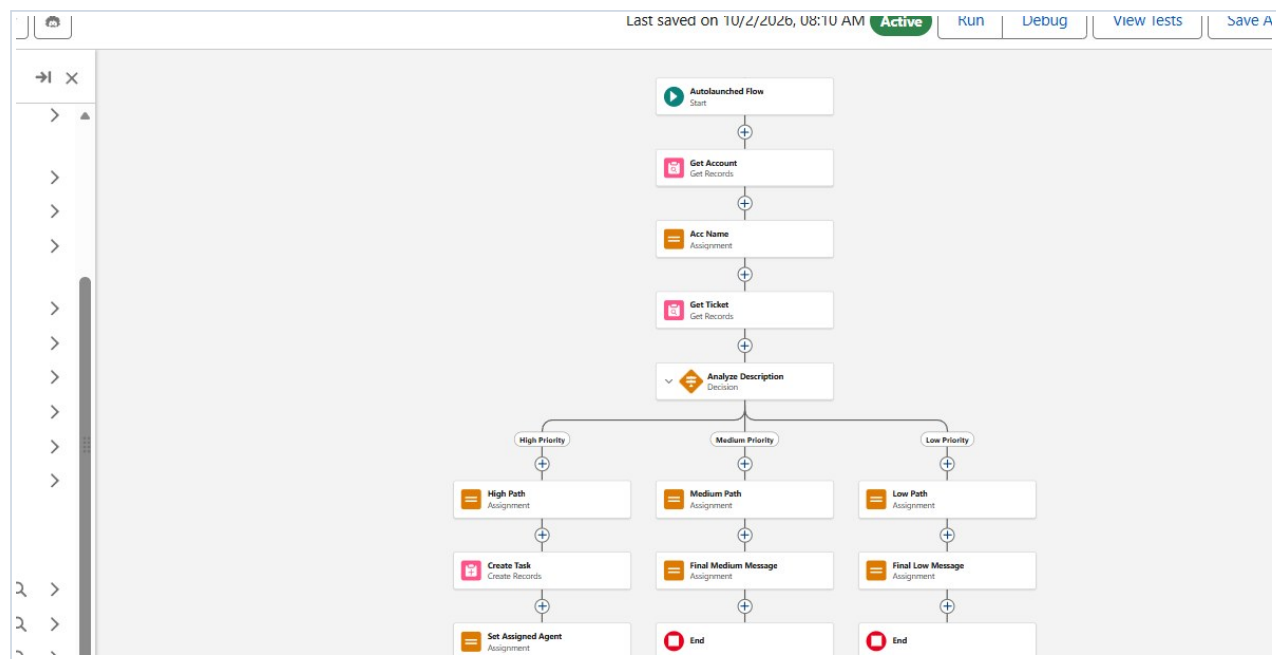


Figure 14.1 – Auto-Launched Flow Builder showing the project automation path (Active)

15. High-Priority Task Automation

When the priority level is High, the Flow follows the High Priority path and creates a Task.

Task Field	Configured Value
Object	Task
Subject	Urgent Ticket Handling
Related To (WhatId)	varTicketId
Priority	High
Status	Not Started
Assigned Support Level	Senior Support Agent (returned in varAssignedTo)

Final Action Messages

Priority	Action Message
High	High priority ticket detected. Assigned to senior agent.
Medium	Ticket marked as medium priority. Will be handled shortly.
Low	Tickets are low priority and queued for processing.

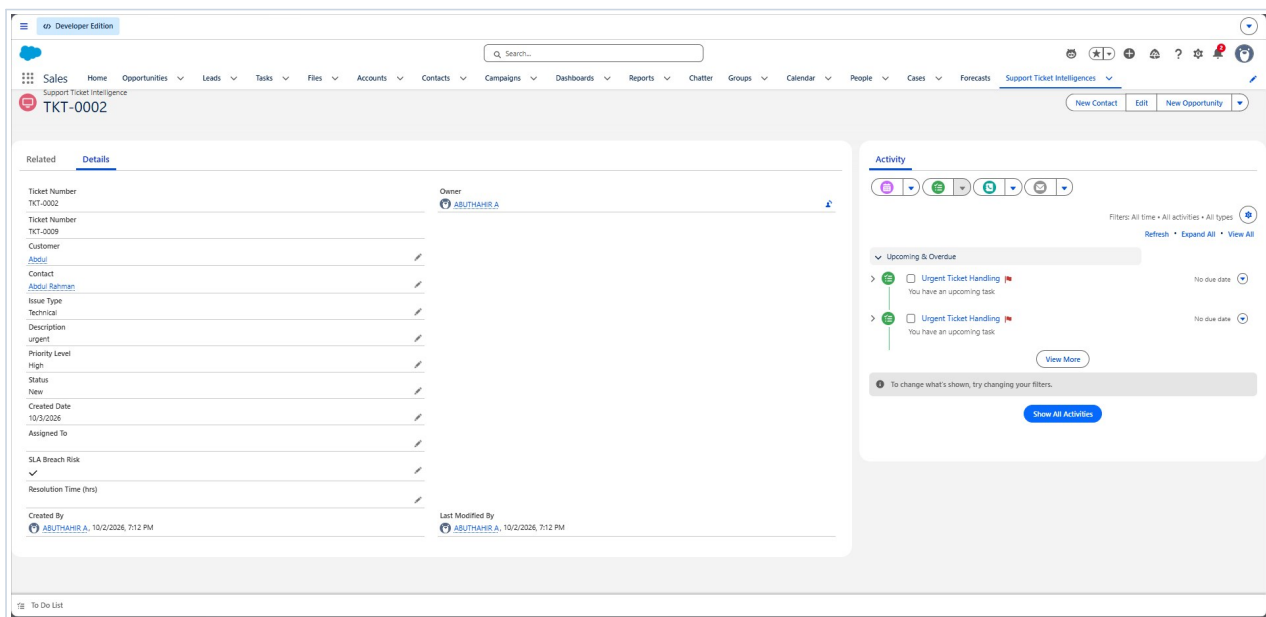


Figure 15.1 – Ticket TKT-0002 with Urgent Ticket Handling tasks created by the Flow

16. Agentforce Subagent Configuration

Agentforce is the conversational layer. A new subagent was added to the Agentforce (Default) agent through Agentforce Builder.

Configuration	Value
Subagent Name	Support Ticket Priority Analysis
API Name	Support_Ticket_Priority_Analysis

Configuration	Value
Classification Description	Analyzes customer support tickets to determine priority based on issue description and urgency; handles requests about ticket status, issue severity and support prioritization.
Scope	Analyze support ticket descriptions, determine the priority level (High, Medium, Low) and trigger backend automation for task assignment.
Out of Scope	Unrelated requests such as billing, subscription management or account updates.

Subagent Instructions

1. Ask the user to provide the Account Name if not already given.
2. Retrieve the latest support ticket associated with the Account.
3. Read and analyze the ticket description.
4. Determine priority: High if it contains “urgent”, “not working”, “failure”; Medium if it contains “issue”, “slow”, “delay”; otherwise Low.
5. Set the priority level accordingly.
6. If the ticket is High priority, trigger the Auto-Launched Flow and ensure a task is created for immediate handling.
7. Assign the ticket to the appropriate support level (for example, senior agent for High priority).
8. Return a clear response: High – urgent handling is initiated; Medium – will be handled shortly; Low – ticket is queued.
9. Do not update records directly or send emails. Use the Flow for all actions.

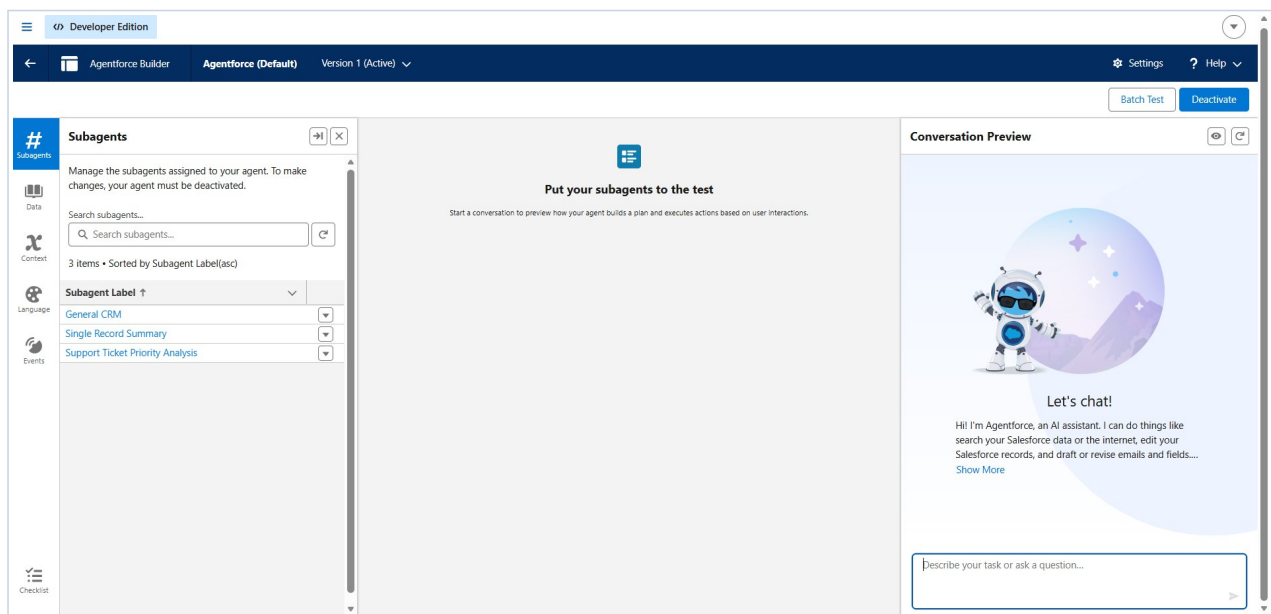


Figure 16.1 – Agentforce Builder showing the Support Ticket Priority Analysis subagent

17. Agentforce Action: Inputs and Outputs

Action: Flow-based action referencing the Auto-Launched Flow. **Description:** analyzes customer support tickets for the provided account name, determines ticket priority from the issue description and triggers automation for high-priority cases. **Loading text:** “Analyzing ticket details and determining priority...”

Input

Variable	Description	Settings
varAccountName	Customer Account Name used to fetch the latest support ticket and analyze its priority.	Text; input required; collected from the user

Outputs

Variable	Purpose
varAccountId	Returns the customer Account identifier.
varActionMessage	Displays the final priority / action message.
varAssignedTo	Indicates the support agent or support level.
varPriorityLevel	Returns High, Medium or Low.
varTicketId	Returns the identifier of the analyzed ticket.

All outputs are text values with “Show in Conversation” enabled.

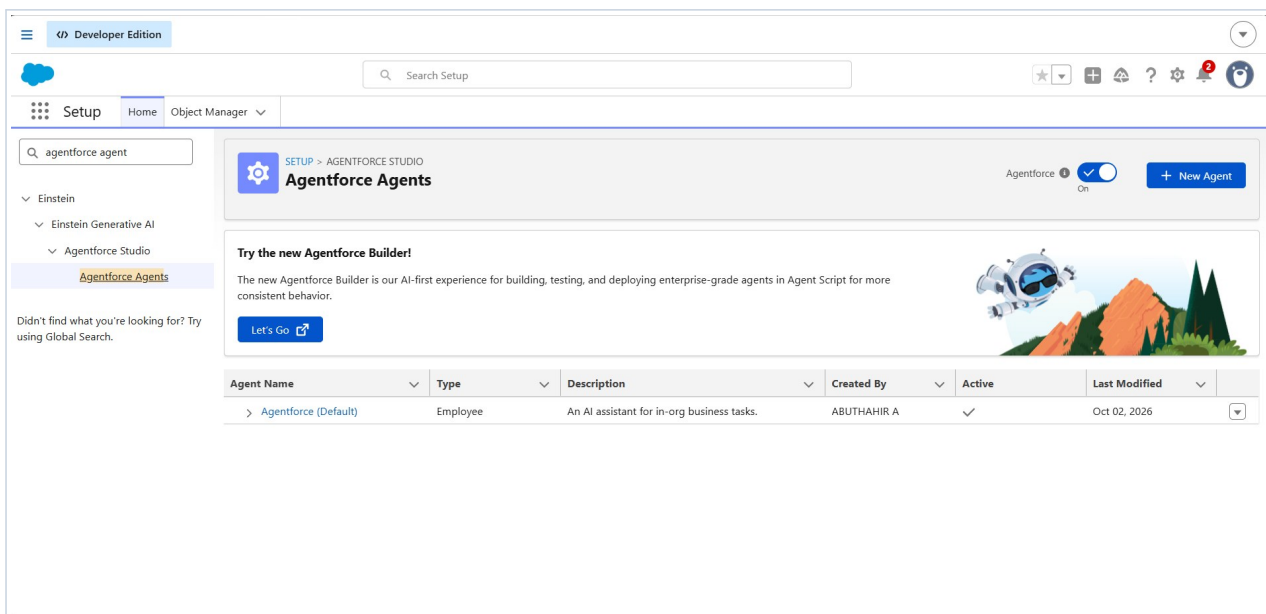


Figure 17.1 – Agentforce Agents page in Salesforce Setup (Agentforce enabled; Agentforce (Default) active)

18. Functional Testing and Verified Result

A live Agentforce conversation was run using the Account Name **Abdul**. The agent retrieved the latest support ticket, analyzed the description and returned the High-priority result and action taken.

Test	Input / Ticket	Expected	Observed Result	Status
TC-1	Account: Abdul; description: "urgent"	High; task created; Senior Support Agent	Message "High priority ticket detected. Assigned to senior agent."; support level Senior Support Agent; priority High; Urgent Ticket Handling task visible in Activity	Verified live
TC-2	Description containing issue / slow / delay	Medium; no task; "will be handled shortly"	Defined in Flow decision and message logic	Configured (live screenshot not included)
TC-3	Description without any keyword	Low; no task; "queued for processing"	Defined as the Flow default outcome	Configured (live screenshot not included)

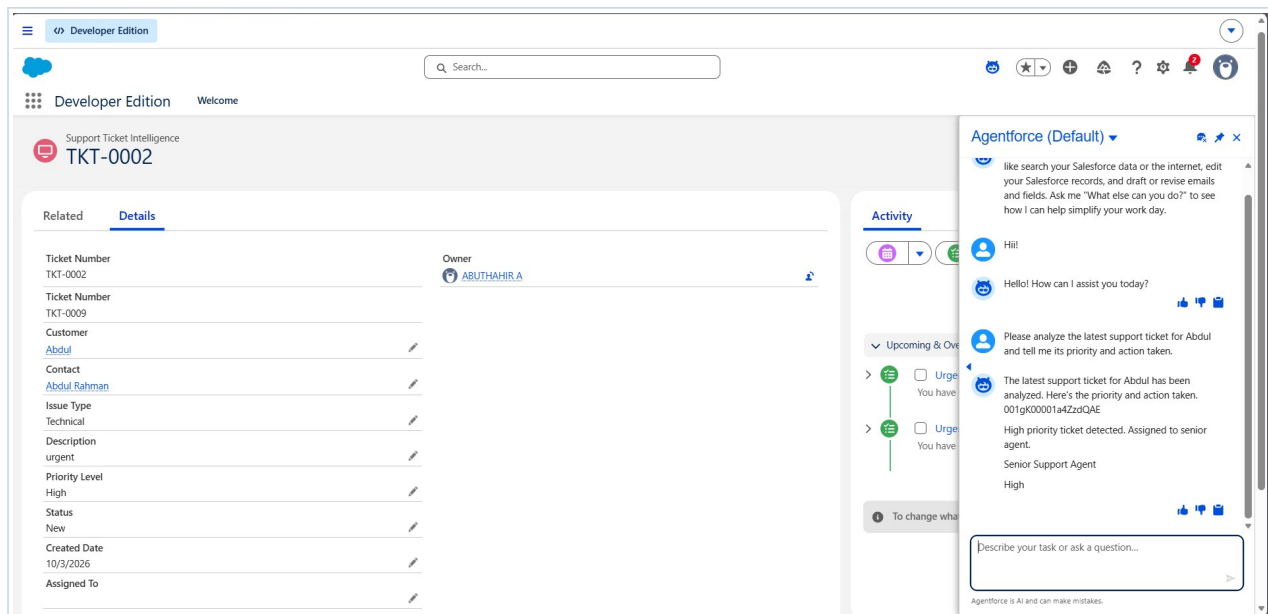


Figure 18.1 – Live Agentforce test result for the Abdul support ticket (TKT-0002)

Observations from the test

- The chat response contains the action message, the support level (Senior Support Agent) and the priority (High), plus one identifier. The identifier shown begins with 001, which is a Salesforce Account Id (varAccountId).
- Two Urgent Ticket Handling tasks appear for TKT-0002. Each run of the High path creates a task, so this is consistent with the Flow having been run twice during testing.
- The Flow returns the computed priority and support level as output variables. In this version it has no Update Records element, so the Priority Level on the record was entered with the test data and the Assigned To lookup was not filled by the Flow.
- The record page shows two fields labelled Ticket Number: the custom field Ticket_Number__c and the standard record Name (both Auto Number), which hold different values (TKT-0002 and TKT-0009). Renaming the label of the standard Name field to "Ticket Name" would remove the ambiguity.

Testing note: the documented live evidence verifies the High-priority path. Medium and Low outcomes are part of the configured Flow logic; this review does not claim a live screenshot for those paths.

19. Advantages

- **Automatic ticket prioritization** – classifies tickets as High, Medium or Low from the description.
- **Faster identification of critical issues** – agents can focus on urgent tickets first.
- **Reduced manual effort** – automates ticket analysis, priority setting, task creation and support-level assignment.
- **Intelligent ticket handling** – Agentforce analyzes the request and starts the backend automation.
- **Automatic urgent task creation** – High-priority tickets create an Urgent Ticket Handling task.
- **Improved support assignment** – High-priority cases get the Senior Support Agent level.
- **Conversational support** – users request ticket analysis by chatting with Agentforce.
- **Better response time** – automation reduces delays in handling critical issues.
- **Scalable operations** – Salesforce automation supports growing ticket volumes.
- **Improved team productivity** – reduces repetitive triage work.

20. Limitations

- **Keyword-based priority logic** – classification depends on keywords such as urgent, not working, failure, issue, slow and delay.
- **Limited understanding of context** – not every urgency signal is captured.
- **Dependence on accurate ticket data** – correct Account, Contact and ticket information is required.
- **Flow and Agentforce configuration dependency** – behaviour depends on the Flow logic and the subagent instructions and action setup.
- **Limited Agentforce scope** – the subagent only handles ticket analysis, priority and backend automation.
- **Priority and assignment are returned as outputs** – the current Flow does not write them back to the ticket record.
- **SLA monitoring is limited** – SLA risk is an optional check, not a complete escalation system (no separate SLA element in Flow V1).
- **Assignment logic may need expansion** – more teams or rules need further configuration.
- **Limited analytics** – priority and resolution analytics are future scope.

21. Future Scope

- Add more priority indicators and business rules, and go beyond simple keyword conditions.
- Update Priority Level and Assigned To on the ticket record from the Flow.
- Add more sophisticated SLA monitoring and escalation.
- Support additional support teams and assignment rules.
- Extend Agentforce outputs with additional ticket information (more variables in the Flow and action).
- Introduce analytics and reports for priority trends, ticket volume and resolution time.

22. Conclusion

The Customer Support Ticket Priority Prediction and Automated Assignment System shows how Salesforce Flow and Agentforce can work together to reduce manual support-ticket handling. The solution retrieves the latest ticket for an Account, analyzes its description, classifies the priority, creates an urgent task for High-priority cases, assigns a support level and returns a clear conversational response.

The project combines structured Salesforce data, deterministic Flow automation and an Agentforce conversational interface. The collected evidence covers the Support Ticket Intelligence object, its fields, the Auto-Launched Flow, the Agentforce configuration, a live High-priority test and the resulting Urgent Ticket Handling task.

Project Completion Summary

Area	Status / Evidence
Requirement Analysis	Completed and documented
Salesforce Data Model	Support Ticket Intelligence object and fields configured (Figure 12.1)
Automation	Auto-Launched Flow active and implemented (Figure 14.1)
Agentforce	Support Ticket Priority Analysis subagent and Flow action configured (Figures 16.1, 17.1)
High-Priority Test	Verified with Abdul / urgent ticket (Figure 18.1)
Task Automation	Urgent Ticket Handling task visible in Salesforce (Figure 15.1)
Project Review	Final Project Review document prepared

23. Appendix

A. Key Project Artifacts

Artifact	Reference
Custom Object	Support Ticket Intelligence
API Name	Support_Ticket_Intelligence__c
Flow	Support_Ticket_Intelligence – V1 (Auto-Launched)
Agentforce Subagent	Support Ticket Priority Analysis
Agentforce API Name	Support_Ticket_Priority_Analysis
High Priority Task	Urgent Ticket Handling
Sample Customer / Ticket	Abdul / TKT-0002

B. Team

Abinesh M • Abuthahir A (Team Lead) • Chundi Jaghava • Joy Abrahamraj • Guru Saran

C. Project Tracker

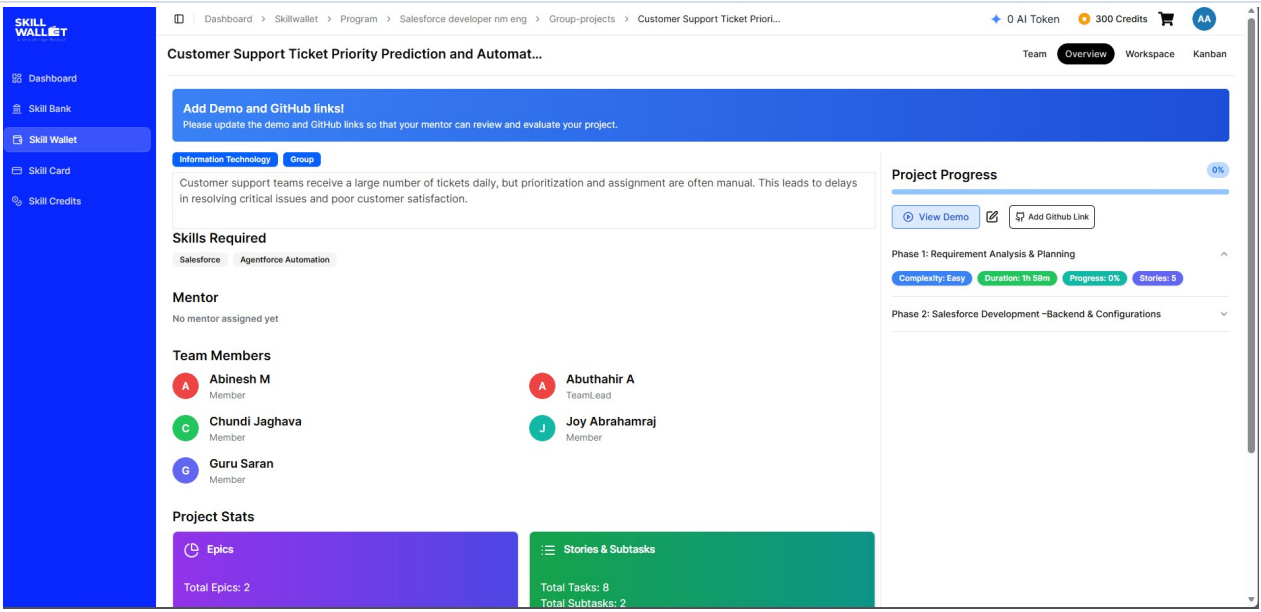


Figure 23.1 – Skill Wallet project workspace overview (team, skills, project stats)